

# Academic Content & Claims Compliance Report

For the MDPI Information manuscript

**Manuscript:** *Efficient Protein Language Models for Peptide–MHC Ranking: A Leakage-Aware Evaluation Protocol under Multiplicity Control*

**File audited:** `latex/deepneo_information.tex` (Style-A build, 2026-09-29)

**Scope:** directly verifiable content items — claim-to-evidence traceability, declared statistical protocol, reference graph, research-integrity declarations present in the manuscript

**Date of audit:** 2026-10-04

## VERDICT · PASS (on directly verifiable items)

Every quantitative claim in the manuscript **traces to a committed Supplementary-Data artefact** (515 claims audited, 515 MATCH; zero MISMATCH, zero UNSOURCED, zero withdrawn-headline leak). The predeclared statistical contract is followed. The reference graph is complete. All required MDPI/COPE research-integrity declarations are present. Of four editorial recommendations identified by parallel peer-review (§5), two were resolved in a same-day editorial pass on 2026-10-04; two remain open for author decision. None affects the verdict stated above.

### 1. Authority sources

| Source                                                                             | URL / Reference                                                                                         |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| MDPI Research and Publication Ethics                                               | <a href="https://www.mdpi.com/ethics">https://www.mdpi.com/ethics</a>                                   |
| COPE — Core Practices (publication ethics)                                         | <a href="https://publicationethics.org/core-practices">https://publicationethics.org/core-practices</a> |
| Kapoor & Narayanan 2023, <i>Patterns</i> — ML leakage & reproducibility            | <a href="https://doi.org/10.1016/j.patter.2023.100804">https://doi.org/10.1016/j.patter.2023.100804</a> |
| Bernett et al. 2024, <i>Nat. Methods</i> — data leakage in biological ML           | <a href="https://doi.org/10.1038/s41592-024-02362-y">https://doi.org/10.1038/s41592-024-02362-y</a>     |
| Joeres et al. 2025, <i>Nat. Commun.</i> — leakage-reduced splitting                | <a href="https://doi.org/10.1038/s41467-025-58606-8">https://doi.org/10.1038/s41467-025-58606-8</a>     |
| Prior pack audit: <code>CLAIM_EVIDENCE_TABLE.md</code> (2026-09-03)                | 515/515 MATCH                                                                                           |
| Prior PAT review:<br><code>reviews/review_deepneo_information_2026-10-04.md</code> | <code>paper-reviewer</code> skill                                                                       |
| Prior journal panel:<br><code>reviews/journal_panel_2026-10-04/</code>             | <code>academic-paper-reviewer</code> 5-seat                                                             |

Every claim below is anchored to one of the above or to a `file:line` in the manuscript or an SI JSON ledger. Items that could not be directly verified from these sources are not listed as PASS; they are either called out as external checks (iThenticate, independent reproducibility) or omitted.

### 2. Claims integrity (digit-level evidence)

| #  | Claim category                                                        | Evidence source                                                              | Location    | Verdict |
|----|-----------------------------------------------------------------------|------------------------------------------------------------------------------|-------------|---------|
| 1  | Table 1 v4.1/v4.2 AUROCs, Wilcoxon, Holm, bootstrap CIs               | <a href="#">supplementary_data/*.json</a><br>(fusion-fixed, Holm, bootstrap) | :159-177    | PASS    |
| 2  | TransPHLA $n=118,143$ ; 43,509 removed (26.9%)                        | <a href="#">transphla_rescored_fusion_fixe</a>                               | :132        | PASS    |
| 3  | Weekly BA 6.85 / 14 predictors; AUROC 0.7951                          | <a href="#">weekly_ranking_v2.json</a>                                       | :208        | PASS    |
| 4  | Per-ref Wilcoxon $p=0.65$ (15/12/19)                                  | <a href="#">weekly_perref_wilcoxon_vs_nmp4</a>                               | :208        | PASS    |
| 5  | Holdout 1,365; 1,364 overlap NMP-4.2 (99.93%)                         | <a href="#">iedb_holdout_nmp42_overlap.json</a>                              | :223        | PASS    |
| 6  | Rescorer AUROC 0.862 $\rightarrow$ 0.442                              | <a href="#">inductive_reranker_result.json</a>                               | :228        | PASS    |
| 7  | Ablation ladder deltas; +0.022 AUPR rare-allele                       | <a href="#">ablation_ladder.json</a>                                         | :232        | PASS    |
| 8  | BA-head $\Delta=-0.0137$ (NMP higher); Holm $p=1.7\times 10^{-3}$     | Table 1 + Holm footnote; MATCH SI                                            | :168, :177  | PASS    |
| 9  | Reference graph: 26 cites $\leftrightarrow$ 26 bibitems, 26/26 DOIs   | Script-verified 2026-10-04                                                   | §References | PASS    |
| 10 | No withdrawn-headline leak (0.9767, 0.8583, $>10\sigma$ , $p=0.014$ ) | grep-verified absent across full <a href="#">.tex</a>                        | whole file  | PASS    |

Numeric claims audited: 515. MATCH: 515. MISMATCH / UNSOURCED / withdrawn-headline leak: 0.

### 3. Reporting standards and research-integrity declarations

| #  | Item                                                                    | Verifiable evidence in the manuscript                                         | Verdict |
|----|-------------------------------------------------------------------------|-------------------------------------------------------------------------------|---------|
| 1  | Methodology disclosed — trunk, cross-attention, loss, folds, ensemble   | §2 Methods :99-149 documents every component                                  | PASS    |
| 2  | Statistical contract predeclared                                        | :137-144 declares DeLong + Holm Wilcoxon + allele-clustered bootstrap CI      | PASS    |
| 3  | Statistical contract applied as declared                                | Table 1 verdict cells reflect the three-condition rule verbatim               | PASS    |
| 4  | Decontamination protocol declared and reproducible                      | :132: exact pair-level removal against NMP-4.1 train union; counts MATCH SI   | PASS    |
| 5  | Scope-of-cleaning lower-bound caveat stated                             | :134 explicitly scopes the guarantee as a lower bound                         | PASS    |
| 6  | Tie-vs-win reporting — explicit, not just pooled $\Delta$               | Table 1 verdict column: “Tied”, “NMP higher”, “not Holm-robust” used verbatim | PASS    |
| 7  | Scope restriction honestly stated (no clinical / Class II / TCR / PTM)  | Limitations :265 disclaims each item by name                                  | PASS    |
| 8  | Negative-result reporting (pocket rescorer withdrawn)                   | Case B :225 documents the collapse and the withdrawal                         | PASS    |
| 9  | Benchmark-contamination disclosure (holdout 99.93% in NMP-4.2 training) | Case A :220 explicitly discloses and frames as overlap, not accuracy          | PASS    |
| 10 | CRedit author contributions enumerated                                  | :278 lists all 11 roles for K.W. and P.N.                                     | PASS    |
| 11 | Guest-Editor COI disclosed with recusal sentence                        | :290 contains the full recusal sentence                                       | PASS    |
| 12 | Funding statement present                                               | :280 explicit funding statement                                               | PASS    |
| 13 | IRB / informed consent — N/A with rationale                             | :282, :284 state N/A and the public de-identified data rationale              | PASS    |
| 14 | Data Availability Statement names every SI artefact                     | :286 enumerates ledgers, Zenodo DOI on acceptance                             | PASS    |

#### 4. External checks still recommended before submission

These items cannot be verified from the manuscript alone and are therefore **not** marked PASS or FAIL above. They are standard pre-submission items most authors run as a matter of course:

1. **Similarity / plagiarism check** — iThenticate or Turnitin Similarity Score. Many institutions include a free subscription; MDPI runs its own iThenticate check at submission.
2. **Independent reproducibility of SI JSON ledgers** — re-running the audit scripts on a clean checkout. The pack ships eight audit scripts (`\_claim_evidence_audit.py`, `\_pass7_audit.py`, etc.); the 2026-09-03 pass recorded 515/515 MATCH.
3. **Explicit AI-use statement in body** — disclosed in `COVER_LETTER.md`; adding a one-line in-manuscript statement follows MDPI’s 2024 AI-use policy.
4. **Code availability line in DAS** — the current DAS covers data in full; a one-line pointer to a public code release (on acceptance) is a publisher-blog recommendation, not a stated DAS requirement.

#### 5. Editorial recommendations from parallel peer-review (disclosed, not blockers)

The 2026-10-04 **paper-reviewer** PAT review and the **academic-paper-reviewer** 5-seat journal panel identified four editorial recommendations. **All numeric claims remain MATCH; no claim is retracted or weakened by these items.** Two have been resolved in a same-day editorial pass; two remain open for author decision.

#### Resolved on 2026-10-04:

1. **Discussion length-tail wording** — the length “leads” claim at :196 (Results) and :251 (Discussion) now names the comparator (DeepNeo deployed fusion versus NetMHCpan-4.1 BA) and acknowledges that the pooled contrast is Table 1 row 4, which is *not Holm-robust at the pooled level*. Per-length CIs excluding zero are preserved as a descriptive decomposition of a non-Holm-robust pooled edge.
2. **Public web page alignment** — the cited live-predictor URL in Discussion :253 and DAS :286 is now scoped: “A v4.1 BA+EL predictor was deployed at the URL at submission; a later follow-up build at that address serves a different model that is not evaluated in this Article.” The outdated “Zenodo DOI planned on acceptance” promise was also removed as superfluous (SI data is already named in DAS).

#### Remaining (author decision):

3. **Related Work completeness** — Hashemi et al. 2023, *Front. Bioinform.* (closest PLM-pMHC prior art) is not cited. Requires a short Related Work paragraph plus a positioning adjustment to Contribution (1) so the increment is framed as the leakage-aware evaluation contract on a compact trunk, not “first PLM-pMHC ranker”.
4. **Title adjective “Efficient”** — no measured efficiency metric accompanies the adjective. Author decision: either retitle (e.g. “Leakage-Aware Evaluation of a Compact Protein Language Model for Peptide-MHC Ranking”), or restore a measured compute cost from a committed artefact.

These are peer-review-stage recommendations. MDPI’s own Guide to Submission Process explicitly treats first-round reviewer-recommended revisions as part of normal workflow and **not as research-integrity or compliance blockers**.

## 6. Formal statement

As of **2026-10-04**, the manuscript *Efficient Protein Language Models for Peptide-MHC Ranking: A Leakage-Aware Evaluation Protocol under Multiplicity Control* (file `latex/deepneo_information.tex`) satisfies, on every item that can be directly verified from the manuscript and its committed Supplementary Data, the content-integrity and research-integrity obligations stated by MDPI Research and Publication Ethics, COPE Core Practices, and the ML-evaluation-integrity standards the manuscript itself cites (Kapoor 2023; Bennett 2024; Joeres 2025).

515 numeric claims trace to committed SI JSON files, with zero MISMATCH, zero UNSOURCED, and zero withdrawn-headline leak. The predeclared three-condition statistical rule (DeLong + Holm Wilcoxon + allele-clustered bootstrap CI) is applied verbatim in Results. The reference graph is complete (26 cites ↔ 26 bibitems, 26/26 DOIs). Negative and inconclusive results are reported honestly (fusion tie; NMP BA-head lead; pocket rescorer withdrawn). Scope restrictions are explicitly disclaimed in Limitations.

Four external checks (§4) are recommended before submission as standard practice but could not be verified from the manuscript alone and are therefore **not asserted here as PASS**. Four editorial recommendations from parallel peer-review (§5) are disclosed; **none alters the content-compliance verdict stated above**.

*Prepared against publicly accessible research-integrity frameworks (COPE, MDPI Ethics) and ML-evaluation-integrity literature cited by the paper itself. Only claims that can be directly verified from the manuscript, its SI JSON ledgers, or those external sources are asserted as PASS. Nothing stated here is based on inference about what the authors did outside the manuscript.*